

Challenge – QGIS Plugin for S2 Cutter for a given city area

Background:

Google had published OpenBuilding dataset <https://sites.research.google/gr/open-buildings/>

The dataset contains 1.8 billion building detections, across an inference area of 58M km² covering Africa, South Asia, South-East Asia, Latin America and the Caribbean.

For each building in this dataset we include the polygon describing its footprint on the ground, a confidence score indicating how sure we are that this is a building, and a pluscode corresponding to the centroid of the building polygon.

Dataset Format:

Building polygons and points are stored in spatially sharded CSVs with one CSV per S2 cell level 4. Each row in the CSV represents one building polygon or point and has the following columns:

latitude: latitude of the building polygon centroid, **longitude**: longitude of the building polygon centroid, **area_in_meters**: area in square meters of the polygon, **confidence**: confidence score [0.65;1.0] assigned by the model, **geometry**: the building polygon in the WKT format (POLYGON or MULTIPOLYGON). This feature is present in only in polygons data, **full_plus_code**: the full plus code at the building polygon centroid.

<https://maps.google.com/pluscodes/>

Sample:

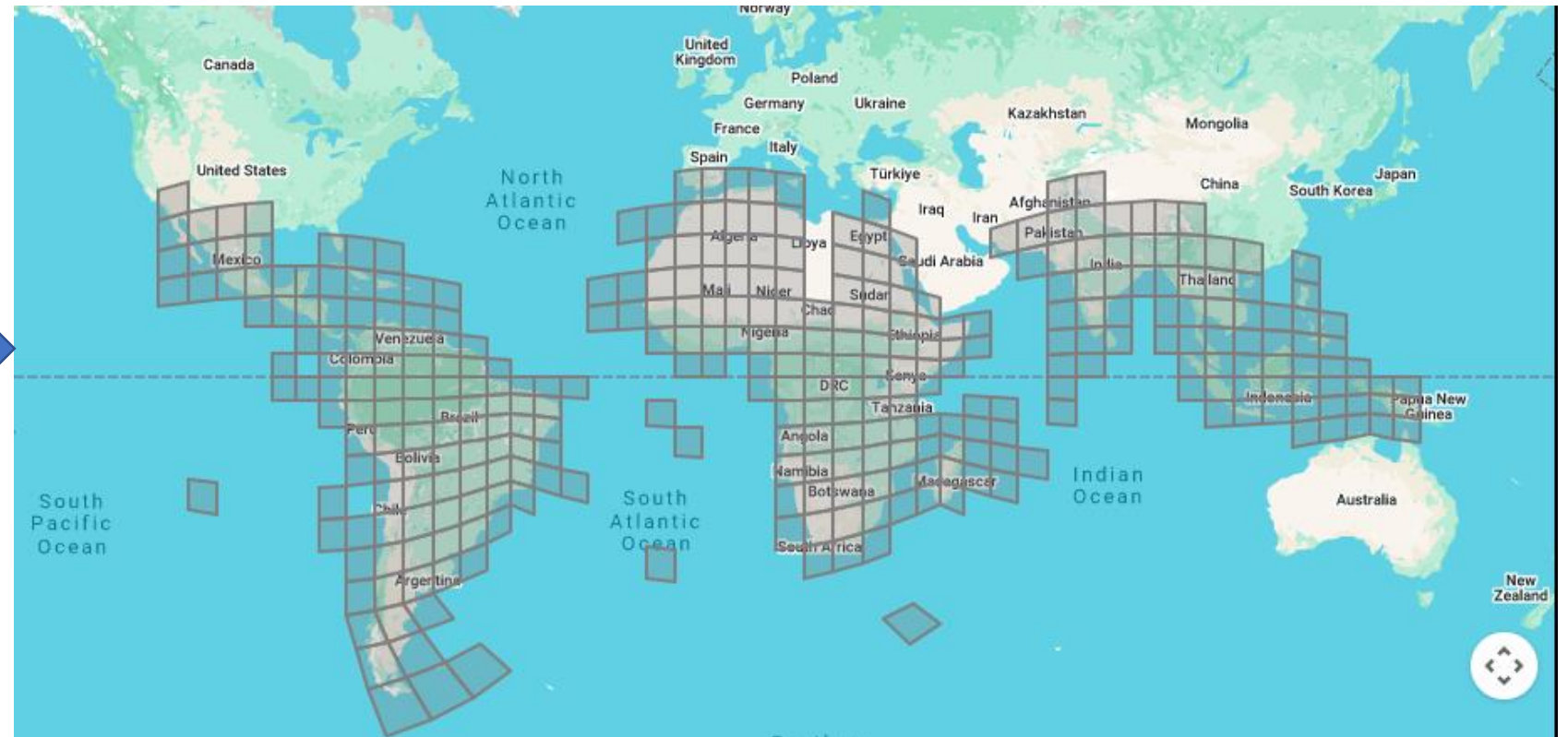
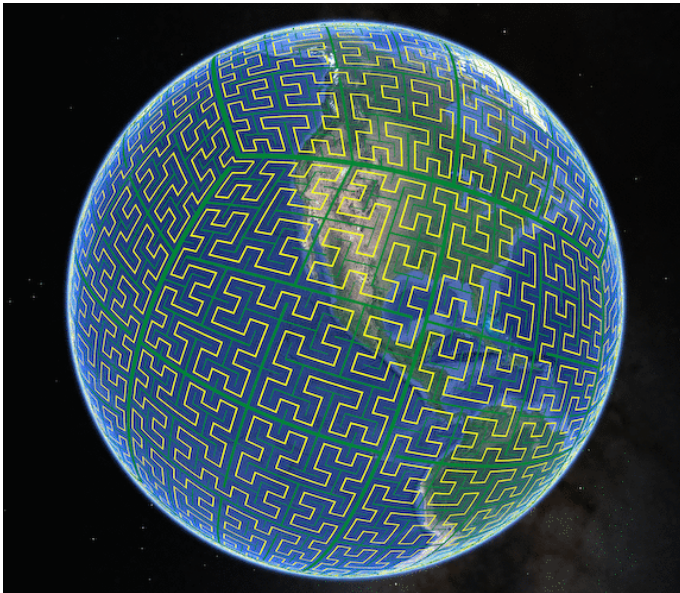
latitude,longitude,area_in_meters,confidence,geometry,full_plus_code

12.97783316,80.22257329,261.7008,0.8421,"POLYGON((80.2226770616936 12.9777750028563, 80.2226754982072 12.9778074983374, 80.2226527873408 12.9778064474131, 80.222648148159 12.9779028678464, 80.2224857607919 12.9778953534553, 80.2224919635459 12.9777664375491, 80.2226770616936 12.9777750028563))",7M42X6HF+42P7

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What is S2 Cell? <https://s2geometry.io/>:

An S2 cell is a cell on a spherical surface and it has geodesic edges. S2 cells are part of a hierarchy dividing up the Earth's surface. They have a maximum of 31 levels, ranging from zero to 30, which define the number of times a cell is subdivided. Levels range from the largest coverage on level zero with area coverage of 85,011,012.19km², to the lowest coverage of 0.44 cm² at level 30.



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What is the Issue?

File size of the dataset for each s2 cell (Level 4) is big – around 1 GB and upto 7 GB (because it covers every building in that area)

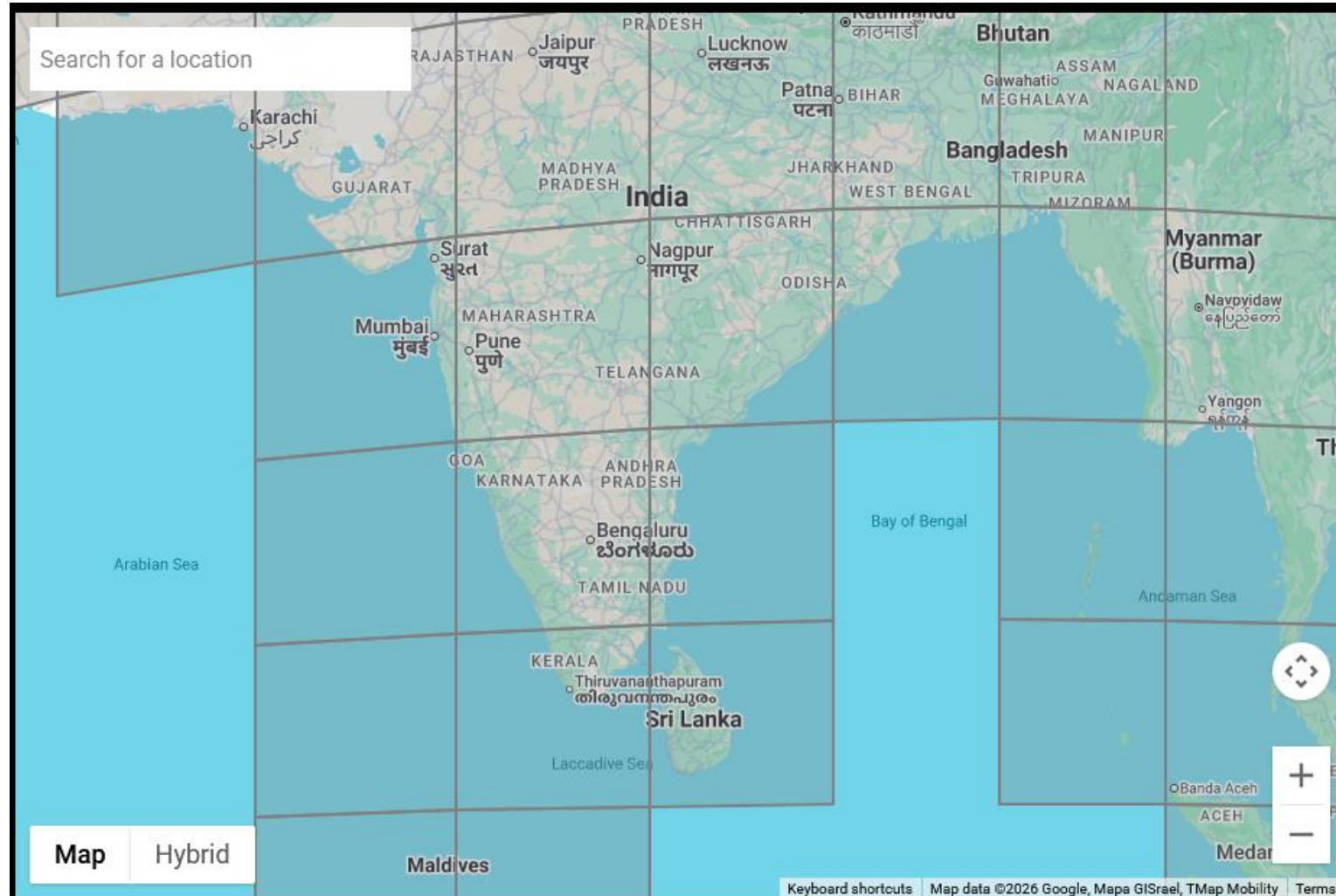
For Example, India is divided into 12 cells.

Big size of dataset demands high compute, Secondly always there is demand for quick and easy analysis.

So it is always preferred to have data for **smaller area covering a city** – Say Hyderabad or Tirupati

Currently, first we have to generate polygon Of interest area using tools like geojson.io

As a second step, chunk the s2Cell data (L4) for this given polygon Using Python libraries



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What is the Task?

Create QGIS plugin (use QGIS 3.44.6)

- ☐ Ability to enter study area (name or lat, long pair) in the field and map zooms to the location.
- ☐ Ability to draw a polygon for a given area
- ☐ Ability to create a new layer with extracted data with the attribute table and csv

What is the Prize?

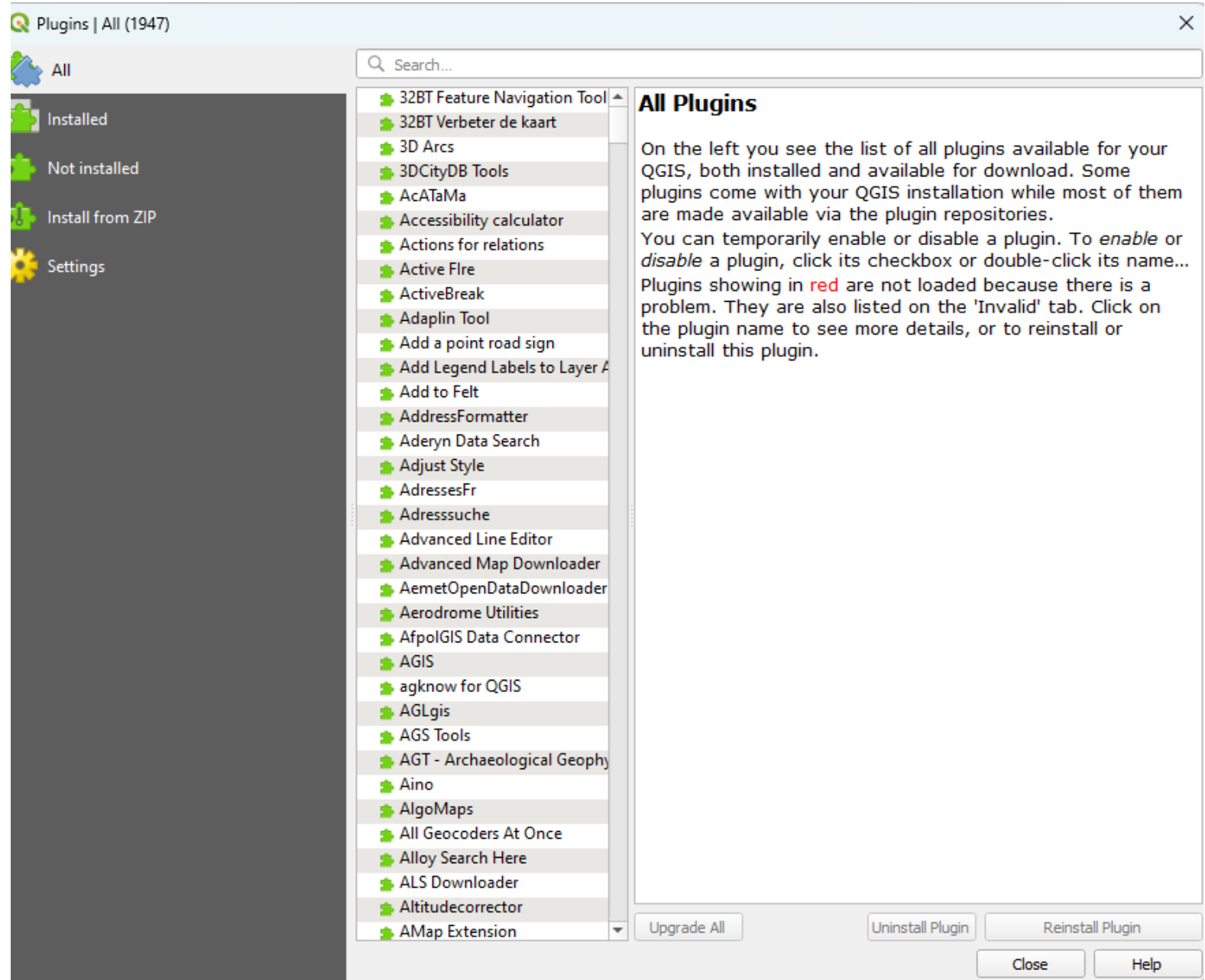
1000 Gems

What is the Timeline?

Complete by **18th Jan**

What is the Submission process?

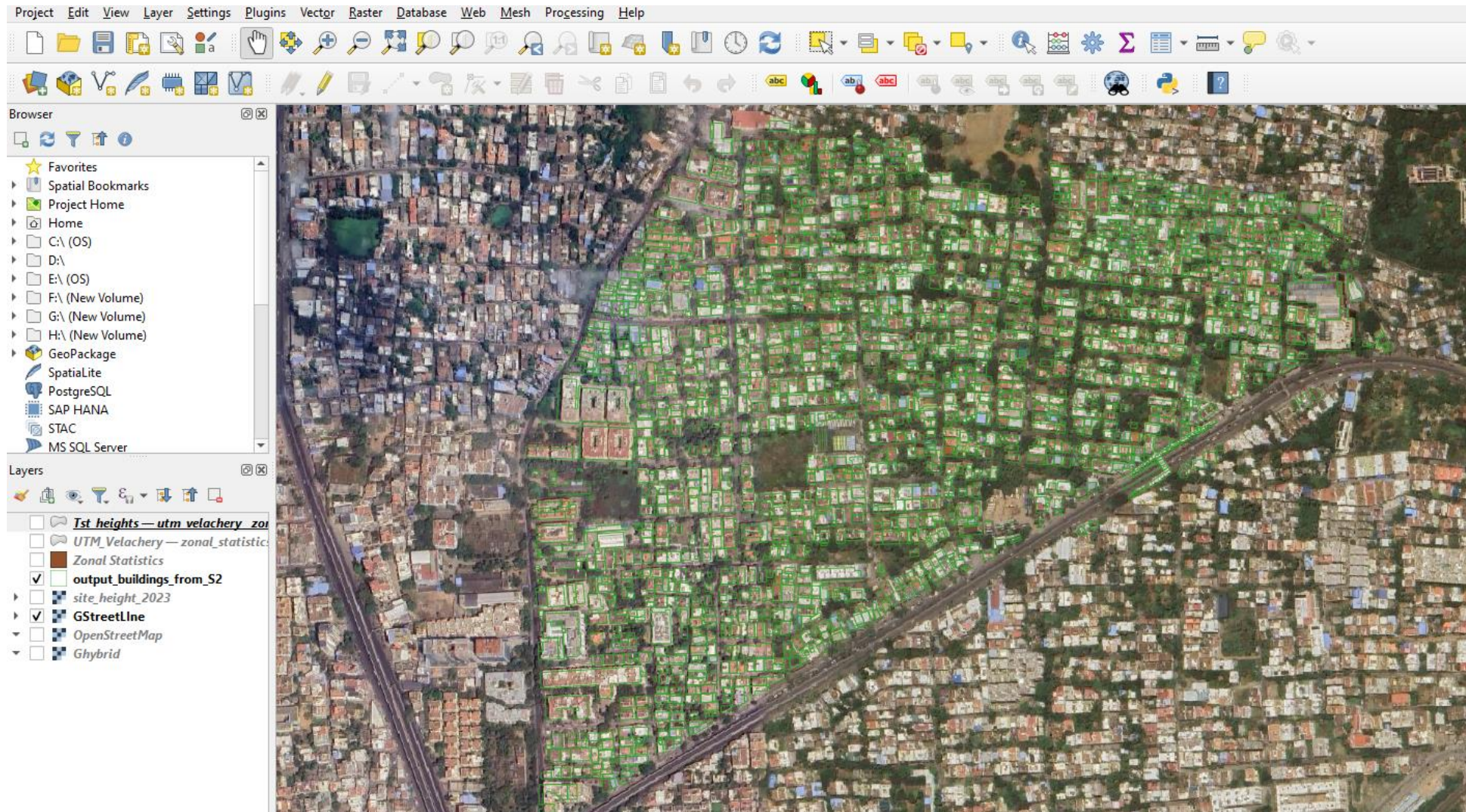
Submit your plugin file (zip)
Upload screenshots
Through MS forms



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Indicative Screenshot expected

This screenshot is for Velachery area, Chennai



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Reference URLs

<https://sites.research.google/gr/open-buildings/>

https://s2geometry.io/devguide/s2cell_hierarchy

https://www.qgistutorials.com/en/docs/3/building_a_python_plugin.html

https://docs.qgis.org/3.40/en/docs/pyqgis_developer_cookbook/plugins/index.html

<https://geojson.io/#map=11.07/13.6558/79.3443>

<https://www.udemy.com/course/qgis-plugin-development-with-python/?srsltid=AfmBOoqMf0MCTB6iQ7jn8haCjLtbL8lf1xLL0r1oYU79QLaSTVohNb9y&couponCode=CP250105G1>

<https://maps.google.com/pluscodes/>

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How this plugin will be useful for our Customers?

Retail – New store location Identification

Using geospatial data, dense areas can be found

Integrating with Google street view image
Demographic pattern can be found

Sustainability – Land Use, Land cover

Using geospatial data, land use, land cover data can be found

Integrating with historical data, usage pattern
Can be analyzed

TATA Projects

For the urban Infrastructure sector,
geospatial data can be useful

Usage pattern- population

CSR

Using geospatial data- lot of potential plans
can be drawn – for example vaccination

Integrating with historical data, usage pattern
Can be analyzed to improve social status of
people

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Sample GOB Chunking code from s2 cell

```
import pandas as pd
from shapely.geometry import Point, Polygon
#(lon, lat)
polygon = Polygon([
    (80.22350044157304, 12.985076115012234),
    (80.22104656472766, 12.981303494742832),
    (80.22124422897912, 12.977841353304896),
    (80.22137681746278, 12.976068216819911),
    (80.23207884483918, 12.982208899398842),
    (80.23126702105696, 12.984045195344152),
    (80.22350044157304, 12.985076115012234)
])

input_file = "3a5_buildings.csv"
output_file = "output_buildings_from_S2.csv"

chunk_size = 100000
header_written = False
```

```
with pd.read_csv(input_file,
    chunksize=chunk_size) as reader:
    for chunk in reader:
        mask = [
            polygon.covers(Point(lon, lat))
            for lon, lat in
zip(chunk["longitude"], chunk["latitude"]
    ])

        filtered_chunk = chunk[mask]

        if not filtered_chunk.empty:
            filtered_chunk.to_csv(
                output_file,
                mode="a",
                index=False,
                header=not header_written
            )
            header_written = True
            print(f"Found {len(filtered_chunk)}
buildings.")
print("Done.")
```